PES UNIVERSITY

Department of Computer Science and Engineering

Software Engineering

LAB 1

Problem Statement #13 – Healthcare & Telemedicine

Patient Health Record Consent Management System

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1. Problem Context & Overview

A patient-centric electronic health data gateway where patients explicitly manage granular, time-bound consent permissions for clinics, diagnostic labs, and consulting doctors to access their medical history.

Target Stakeholders / Actors: Patient, Clinic Administrator

2. Requirements Table

The table below contains exactly five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001 and NFR-002).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow patients to grant time-bounded (e.g., 24-hour) access permissions for specific diagnostic records to verified clinic doctors.	High	Pass: Clinic access automatically revokes upon expiry timestamp. Fail: Clinic accesses records after consent expiration.	Gives patients control over who can access their health records and for how long.	Consent expiry should be clearly shown.
FR-002	Functional	The system shall allow patients to revoke an active consent before its	High	Pass: After revocation, the selected doctor or	Patients should be able to stop access whenever they want.	Revocation should take effect immediately.

		expiry time.		clinic cannot access the records. Fail: Access is still possible after revocation.		
FR-003	Functional	The system shall allow verified doctors and authorized healthcare providers to access only the health records for which valid consent has been granted.	High	Pass: Only records covered by an active consent are displayed. Fail: Records without valid consent are displayed.	Prevents unauthorized access to patient health information.	Consent must be checked before access.
FR-004	Functional	The system shall allow the clinic administrator to verify and manage the authorization status of doctors requesting patient records.	High	Pass: Only verified doctors are marked as authorized to access records. Fail: An unverified doctor is allowed to access records.	Ensures that only authorized healthcare personnel receive access.	Verification status should be maintained by the system.
FR-005	Functional	The system shall allow patients to view their active and expired consent permissions.	Medium	Pass: Patient can see provider, records covered, grant time and expiry time for each consent. Fail: An existing consent is missing.	Helps patients understand and control their record sharing.	Expired permissions should be clearly identified.
NFR-001	Nonfunctional – Performance & Security	All consent grant, revoke, and access events shall be permanently written to an append-only audit trail.	High	Pass: Consent and access events are recorded and cannot be modified or deleted through normal system operations. Fail: An event is missing or can be altered.	Provides accountability and helps track access to sensitive health information.	Audit records should contain event and time.
NFR-002	Nonfunctional – Security	The system shall prevent unauthorized users from accessing patient health records and consent information.	High	Pass: An unauthorized access attempt is denied. Fail: An unauthorized user can view patient records or consent information.	Patient health information must be protected from unauthorized access.	Access control should be checked before records are displayed.